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SciScore Report

Below you will find your SciScore report containing three tables. Your score is calculated based on adherence to scientific rigor criteria (Table 1) and identification of key biological resources (Table 2). Table 3 contains statistical tests and oligonucleotides but is not scored. If SciScore makes any mistakes, please contact us to help us learn and improve.

Table 1: Rigor Adherence Table

Ethics
Consent: The cells were then washed thrice with PBA and taken for fluorescence-activated cell sorting (FACS) analysis . CD4+ T cell isolation from blood After consent from healthy donors , blood was collected in preservative-free anticoagulant (0.2 final concentration of EDTA) and processed immediately for peripheral blood mononuclear cell (PBMC) isolation .
IRB: Ethics statement The study involving peripheral blood cells from healthy donors was approved by the institutional ethics committee.
Inclusion and Exclusion Criteria
not detected.
Attrition
not detected.
Sex as a biological variable
not detected.
Subject Demographics
Age: not detected.
Weight: not detected.
Randomization
not detected.
Blinding
Investigators were blinded to the allocation of small molecules during experiments and outcome assessment for checking the robustness and reproducibility of the phenotypes.
Power Analysis
Data reporting No statistical methods were used to predetermine sample size.
Replication

Number: The secondary screen was performed in at least triplicate.	
Cell Line Authentication	
not detected.	
Data Information	
Availability: The deep sequencing data generated in this study are uploaded to the National Center for Biotechnology Information (NCBI) and can be accessed with ID Bio project PRJNA793002.	
Identifiers: The data are available as Bio project PRJNA793002.	PRJNA793002

Table 2: Key Resources Table

Your Sentences	REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies			
Isolated CD4+ T cells were characterized using FACS after counter-staining with anti-CD4-APC antibody (1:20 in PBA) (130-113-812 - CD4-APC , human Miltenyi Biotec) and anti-CD3-FITC labeled antibody .	anti-CD4-APC		
	anti-CD3-FITC		
Next , cells were collected by centrifugation at 500 x g for 5 min and incubated on ice with a primary CXCR4 antibody (1:500 dilution in PBA) for 3 h .	CXCR4		
Blocking of membranes was carried out by incubation with the Odyssey Blocking Buffer (LI-COR Biosciences) for 20-30 min , followed by both primary and secondary antibody incubations for 1 h each at room temperature , each of which was followed by three washes of 5 min each . Detection of Cas9 and b-actin was carried out using mouse anti-Cas9 antibody (Clontech) and rabbit anti-beta actin antibody (LI-COR Biosciences) , respectively .	anti-Cas9		
	anti-beta actin		
Secondary antibodies used were either IR dye 680 goat anti-mouse or IR dye 800 goat anti-rabbit (LI-COR Biosciences) .	anti-mouse		
	anti-rabbit		
Blocking of the membrane was carried out by incubation with the Blocking Buffer (Bio-Rad) for 30 min at room temperature, followed by both primary (AF3025-SP Human TGF-bRIALK5 Antibody Novusbio) and secondary antibody (IRDYe 680 Donkey anti-Goat Thermo Fisher Scientific) incubations for 1 h each at room temperature, each of which was followed by three washes of 5 min each.	anti-Goat		
Experimental Models: Cell Lines			
Materials and methods Cell culture and reagents HeLa, HT1080, MCF7, and A549 were obtained from the American Type Culture Collection (	MCF7		Suggestion: RRID:CVCL_0031(link)
HEK293T (ECACC) and JurkatTAg (JTAg) were reported earlier in Rosa et al. 23 K562 and SW480 were	K562		Suggestion: RRID:CVCL_0004(link)

obtained from National Centre for Cell Science (NCCS)			
HEK293T , HeLa , K562 , HT1080 , SW480 , and A549 cells were cultured in DMEM (Gibco) supplemented with 10 fetal bovine serum (FBS U.S. origin , certified , heat-inactivated serum Gibco) , 2 mM GlutaMAX , and 1x penicillin-streptomycin (Gibco) .	HEK293T		Suggestion: RRID:CVCL_0063(link)
	SW480		Suggestion: RRID:CVCL_0546(link)
	A549		Suggestion: RRID:CVCL_0023(link)
The A549 , HeLa , HEK293T , HT1080 , K562 , and JTAg luciferase cell lines were created using lentiviral infection (MOI = 0.1) , and the transduced population was selected with puromycin (Invivogen) .	HT1080		Suggestion: RRID:CVCL_0317(link)
TGF-bRI-knockout cell generation and western blot For the generation of TGF-bRI-knockout cells, HEK293T and HeLa cells were transduced with lentiviral vectors carrying guide RNA against TGF-bRI in the presence of Repsox (10 mM).	HeLa		Suggestion: RRID:CVCL_0030(link)
Recombinant DNA			
A lentiviral pScalps mGFP puro vector was constructed for generating stable cell lines for loss-of-function editing assays .	pScalps mGFP puro		
For this , an mGFP encoding sequence was released from the parental mGFP plasmid with NheINotI and cloned into a lentiviral mammalian expression vector pScalps_Puro (a kind gift from Silvia Monticelli plasmid 99636 Addgene) with XbaINotI sites .	mGFP		Suggestion: RRID:Addgene_62231(link)
	pScalps_Puro		Suggestion: RRID:Addgene_99636(link)
pScalps-eSpcas9-Zsgreen plasmid was further modified to include a U6-driven gRNA expression cassette.	pScalps-eSpcas9-Zsgreen		
The guide RNAs were cloned either at BsmBI or BbsI sites in pLentiCRISPR-E or pX458 or pScalps-eSpCas9-Zsgreen (48138 Addgene) .	pLentiCRISPR-E		Suggestion: RRID:Addgene_78852(link)
	pX458		
For editing the SERINC5 locus in various cells , pX458DCas9 gRNA SERINC5 was generated by removing the SpCas9-GFP by XbaINotI and used with Cas9-expressing plasmid.	Cas9-expressing		
The amplified product was cloned in pTZ57R using HindIII and EcoRI .	pTZ57R		
At 50 confluency , cells were transfected (at T = -12 h) using the calcium phosphate method with pLentiCRISPR E (200 ng) , pX458	pCDB-GFP		
	pTAG-RFP657		

gRNA GFP DCas9 (200 ng) (target gRNA)pX458 gRNA FLucDCas9 (control gRNA) (200 ng) , and pCDB-GFP N1 (50 ng) pTAG-RFP657 (50 ng) (transfection control) .			
psPAX2 (6 mg) , and pMD2 . g (2 mg) were co-transfected in HEK293T cells using the calcium phosphate method in a 10 cm2 cell culture dish , seeded 1 day before transfection .	psPAX2		
	pMD2		Suggestion: RRID:Addgene_12259(link)
HIV-1 virus production , CD4+ primary T cell infection , and p24 staining To infect CD4+ primary T cells , we generated HIV-1 particles using HEK293T as a virus producer cells by co-transfecting 7 mg pNL4-3 Env-Nef- and 1 mg pHXB2 X4-tropic envelope 48 in a 10 cm2 cell culture dish using the calcium phosphate transfection method.	pNL4-3		
SMAD inhibition reporter assay To check the effect of Repsox on the inhibition of the TGF-bRI pathway , we used the SMAD reporter plasmid SBE4-Luc (16495 Addgene) .	SBE4-Luc		
Next , cells were transfected with SBE4-Luc , pRL-TK , along with Cas9 and gGFPgRFP plasmids .	pRL-TK		
	gGFPgRFP		
Cas9 VLPs production and target cells transduction To produce virus-like particles carrying Cas9 and guide RNA , HEK293T cells were seeded in a 10 cm2 plate and co-transfected with pcDNA-Cas9 (10 mg) , pTZ tRNA (glutamine) gGFPgFluc (10 mg) , and pMD2 .	pcDNA-Cas9		
Software and Algorithms			
Sample acquisition was done using FACS Aria III (BD Biosciences) , and the data were analyzed using FlowJo software (FlowJo) .	FlowJo		Suggestion: FlowJo, RRID:SCR_008520(link)
For the analysis of CXCR4 editing in JTAg cells , 1 x 107 cells were electroporated using Bio-Rad Gene PulserXcell with eSpCas9 (5 mg) , px458 gCXCR4 DCas9gFLuc DCas9 (5 mg) .	Bio-Rad Gene PulserXcell		
	Gene		
Software and statistical analysis The significance of results was statistically analyzed using Students unpaired t test using GraphPad Prism 9.	GraphPad Prism		

Specific portions of images were produced using BioRender.	BioRender		

Other Entities Detected

Your Sentences	Recognized Entity
Statistical Tests	
Software and statistical analysis The significance of results was statistically analyzed using Students unpaired t test using GraphPad Prism 9.	Students unpaired t test
Oligonucleotides	
The reduction in the half-life of the GFP was achieved by fusing it with a CDB nucleotide sequence , which was amplified from JTAg cDNA using forward GCCACCATGGCGCTCCGAGTCACC AG and reverse TCTTTAACAGGCTCAGGTTC primers .	GCCACCATGGCGCTCCGAGTCACC AG
	TCTTTAACAGGCTCAGGTTC
Unresolved RRIDs	
For this , an mGFP encoding sequence was released from the parental mGFP plasmid with NheINotI and cloned into a lentiviral mammalian expression vector pScalps_Puro (a kind gift from Silvia Monticelli plasmid 99636 Addgene) with XbaINotI sites .	plasmid 99636

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